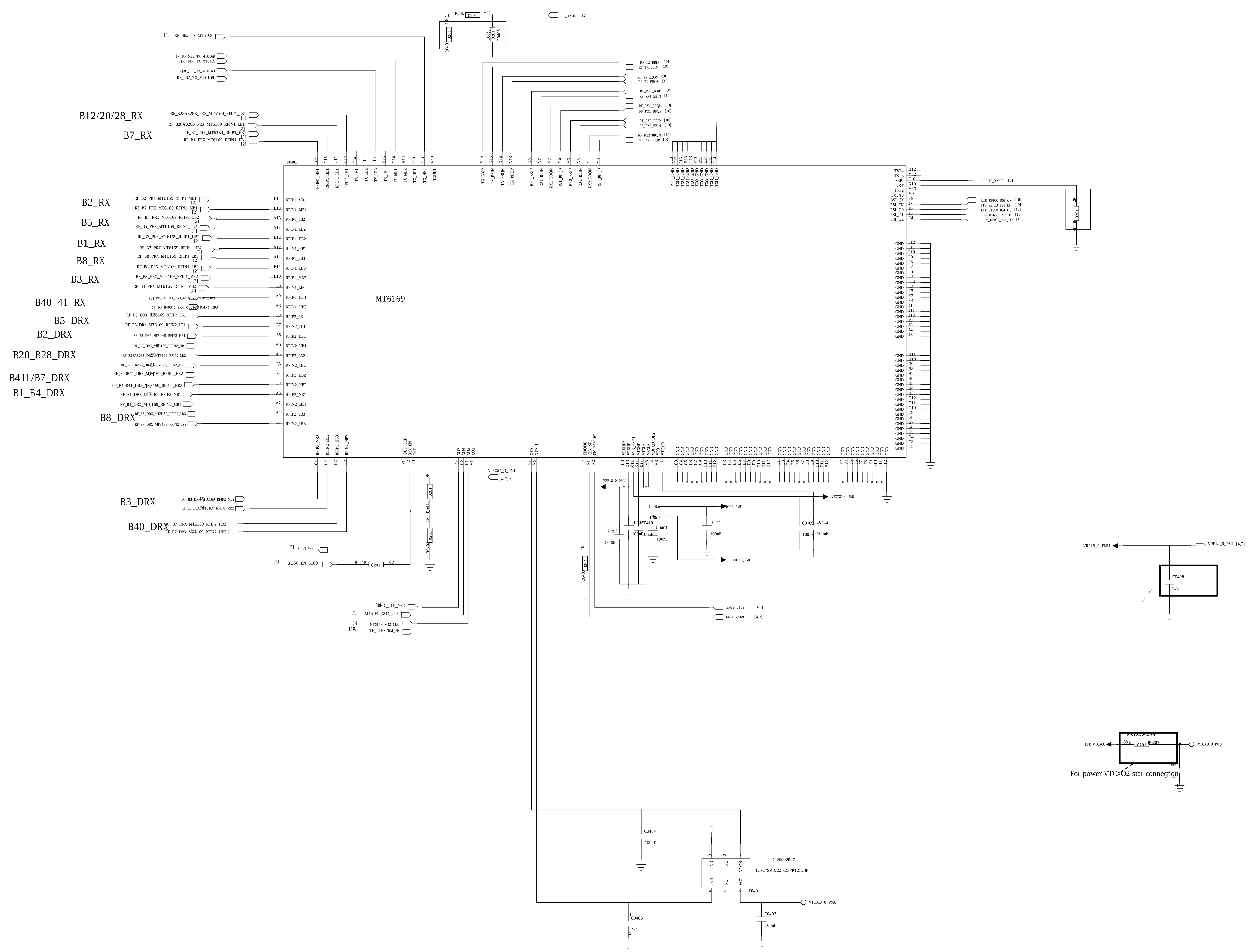
COMPANY:	Shanghai Longcheer
TITLE:	L3400
DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>
CODE: <Code>	SIZE: A0
DRAWING NO: <Drawing Number>	REV: A1
SCALE: <Scale>	SHEET: 1 of 19


CMCC 3M:11522176 4M:11522178
B38 & B40 PRX
B8 TRX
1.8*1.4
2G_PAIN_HB
2G_PAIN_LB
MT6169 TX Ouput need a DC block(MUST).
Pi attenuation for LB BPSK TX in RX band noise
I2C Address:0x44H
B1 TRX
1.8*1.4
B4 TRX
1.8*1.4
B2 TRX
1.8*1.4
B3 TRX
1.8*1.4
B5 TRX
1.8*1.4
B7 TRX
1.8*1.4
B28A------8531
B28A TRX
1.8*1.4
B28B TRX
1.8*1.4
RF_ANT_TRX_TXM_CARKIT
RF_ANT_TRX_TXM_ASM
MIPI1_SDATA
MIPI1_SCLK
RF_B40B41_PRX_PA
RF_B8_TRX_DPX
RF_B8_PRX_DPX
RF_B1_TRX_DPX
RF_B1_PRX_DPX
RF_B4_TRX_DPX
RF_B4_PRX_DPX
RF_B2_TRX_DPX
RF_B2_PRX_DPX
RF_B3_TRX_DPX
RF_B3_PRX_DPX
RF_B5_TRX_DPX
RF_B5_PRX_DPX
RF_B7_TRX_DPX
RF_B7_PRX_DPX
RF_B28A_TRX_DPX
RF_B28A_PRX_DPX
RF_B28B_TRX_DPX
RF_B28B_PRX_DPX
RF_B12B28A28B_PRX_DPX
Shanghai Longcheer
L3400
<Drawn By>
<Drawn Date>
<Checked By>
<Checked Date>
<QC By>
<QC Date>
<Released By>
<Release Date>
<Code>
A0
<Drawing Number>
A1
<Scale>
19


CMCC 3M:11548115 4M:11548117
L352/L353 share pad
share pad
RF_ANT_DRX_CARKIT
RF_ANT_DRX_ASM
B20/28AB DRX
RF_B28_DRX_SAW_IN
RF_B12B28AB28B_DRX_SAW_OUT
B7 DRX
RF_B7_DRX_SAW_IN
RF_B7_DRX_SAW_OUT
B1/B8 DRX
RF_B1B8_DRX_SAW_IN
RF_B8_DRX_SAW_OUT
RF_B1_DRX_SAW_OUT
B2/B5_DRX
RF_B2B5_DRX_SAW_IN
RF_B5_DRX_SAW_OUT
RF_B2_DRX_SAW_OUT
B3
1.1*0.9
RF_B3_DRX_SAW_IN
RF_B3_DRX_SAW_OUT
RF_B38B40B41_DRX_SAW_OUT
RF_B38B40B41_SWITCH_OUT
RF_B38B41_DRX_SAW_IN
RF_B38B41_DRX_SAW_OUT
B38/41
B38/B41B(2555~2655MHz)_DRX
B40 DRX
RF_B40_DRX_SAW_IN
RF_B40_DRX_SAW_OUT
1.1*0.9
COMPANY: Shanghai Longcheer
TITLE: L3400
<Drawn By>
<Drawn Date>
<Checked By>
<Checked Date>
<QC By>
<QC Date>
<Released By>
<Release Date>
CODE: <Code>
SIZE: A0
DRAWING NO: <Drawing Number>
REV: A1
SCALE: <Scale>
SHEET: 3 of 19

MT6169



REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:

COMPANY:	Shanghai Longcheer		
TITLE:	L3400		
DRAWN: <Drawn By>	DATED: <Drawn Date>		
CHECKED: <Checked By>	DATED: <Checked Date>		
QUALITY CONTROL: <QC By>	DATED: <QC Date>		
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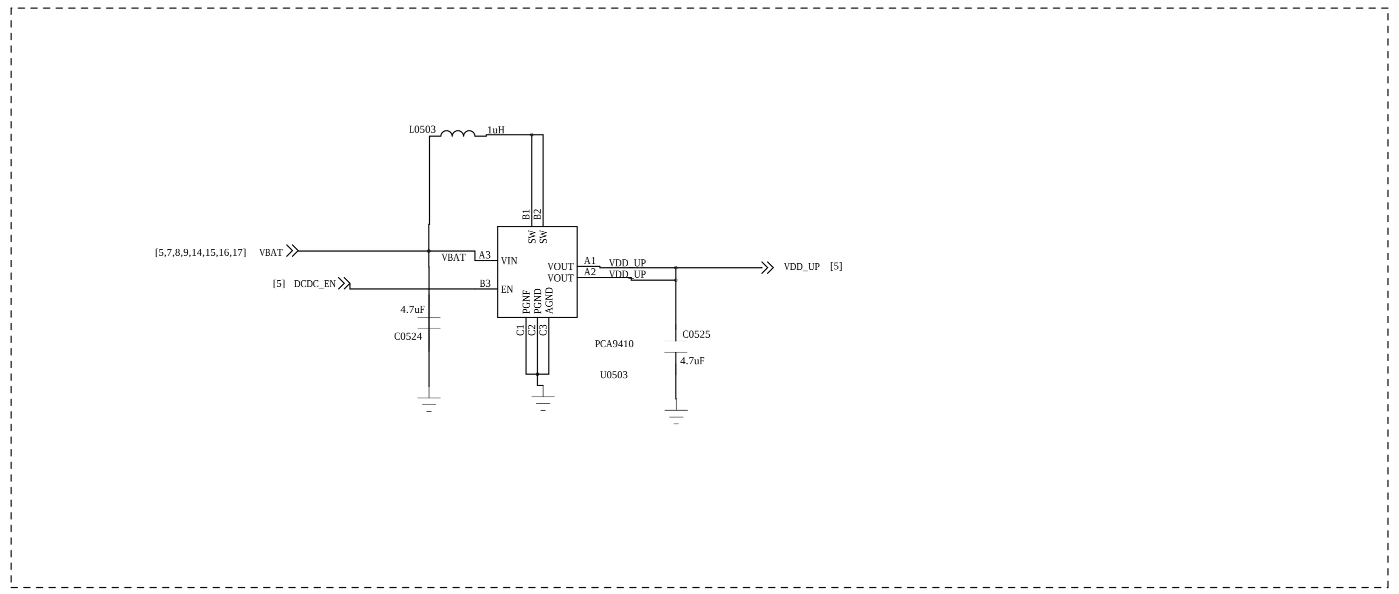
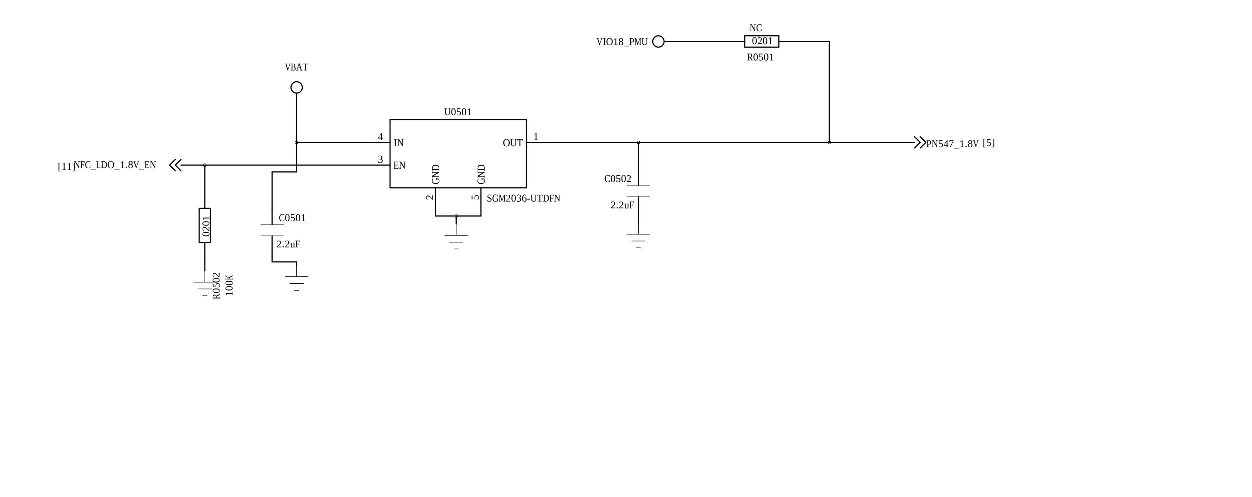
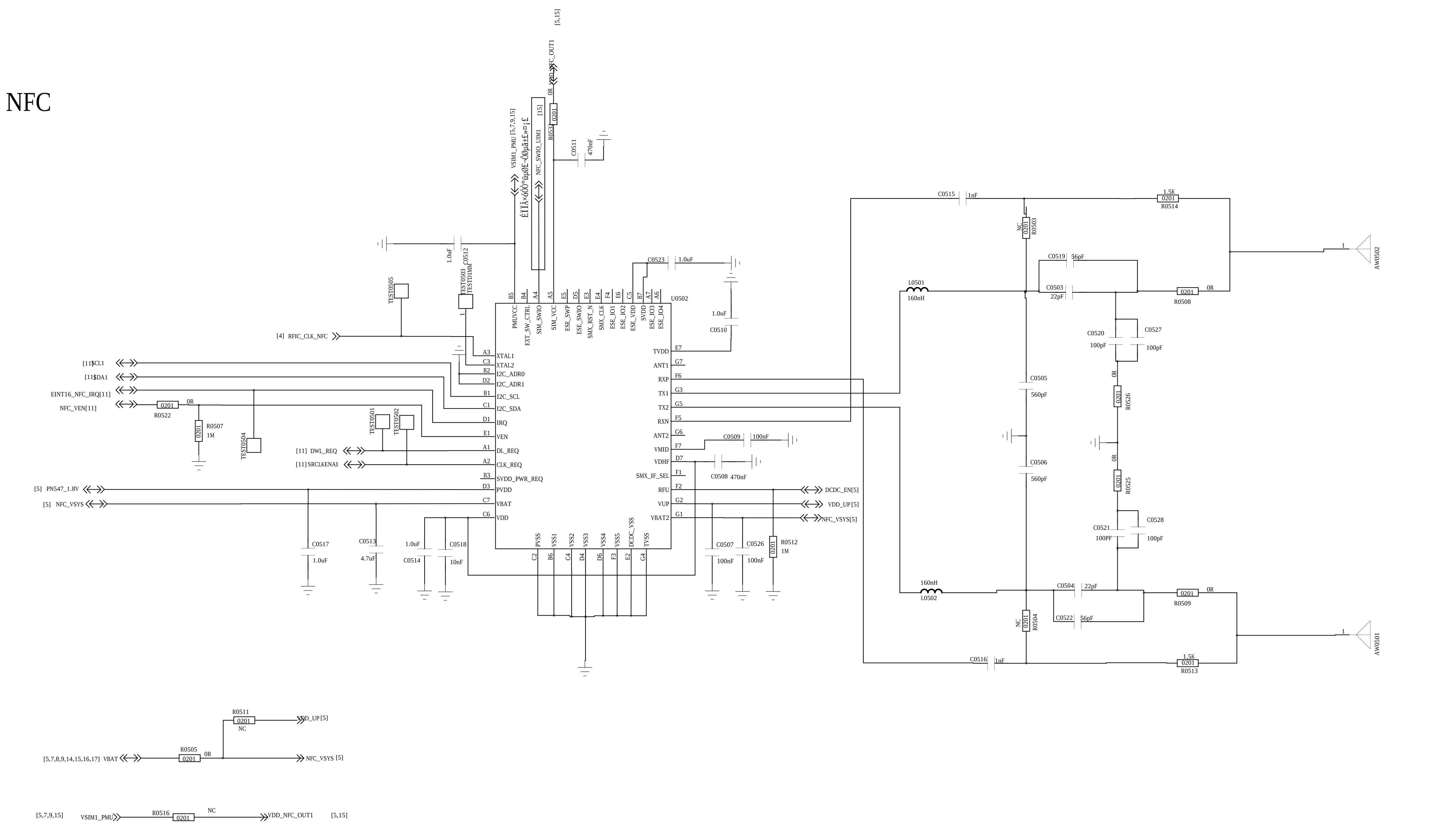
SCALE: <Scale>		SHEET: 4 of 19	



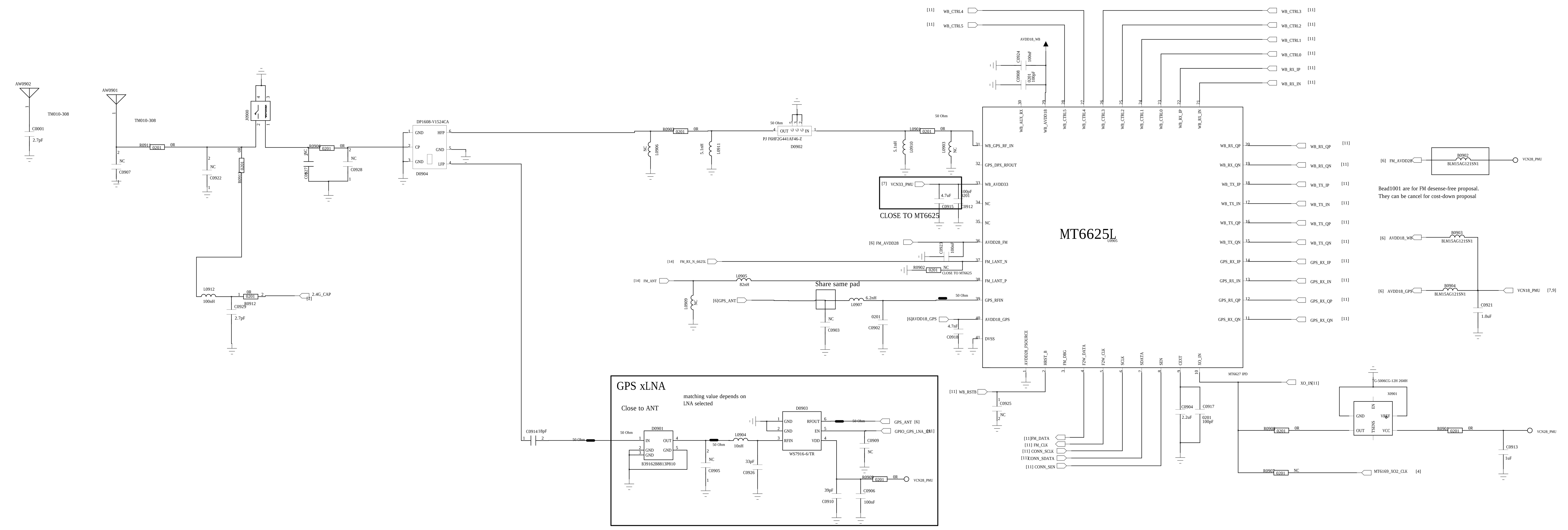
NFC power



NFC



I2C_ADR1	I2C_ADR0	WRITE	READ
0	0	0X50	0X51
0	1	0X52	0X53
1	0	0X54	0X55
1	1	0X56	0X57



REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

COMPANY: Shanghai Longcheer

TITLE: L3400

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
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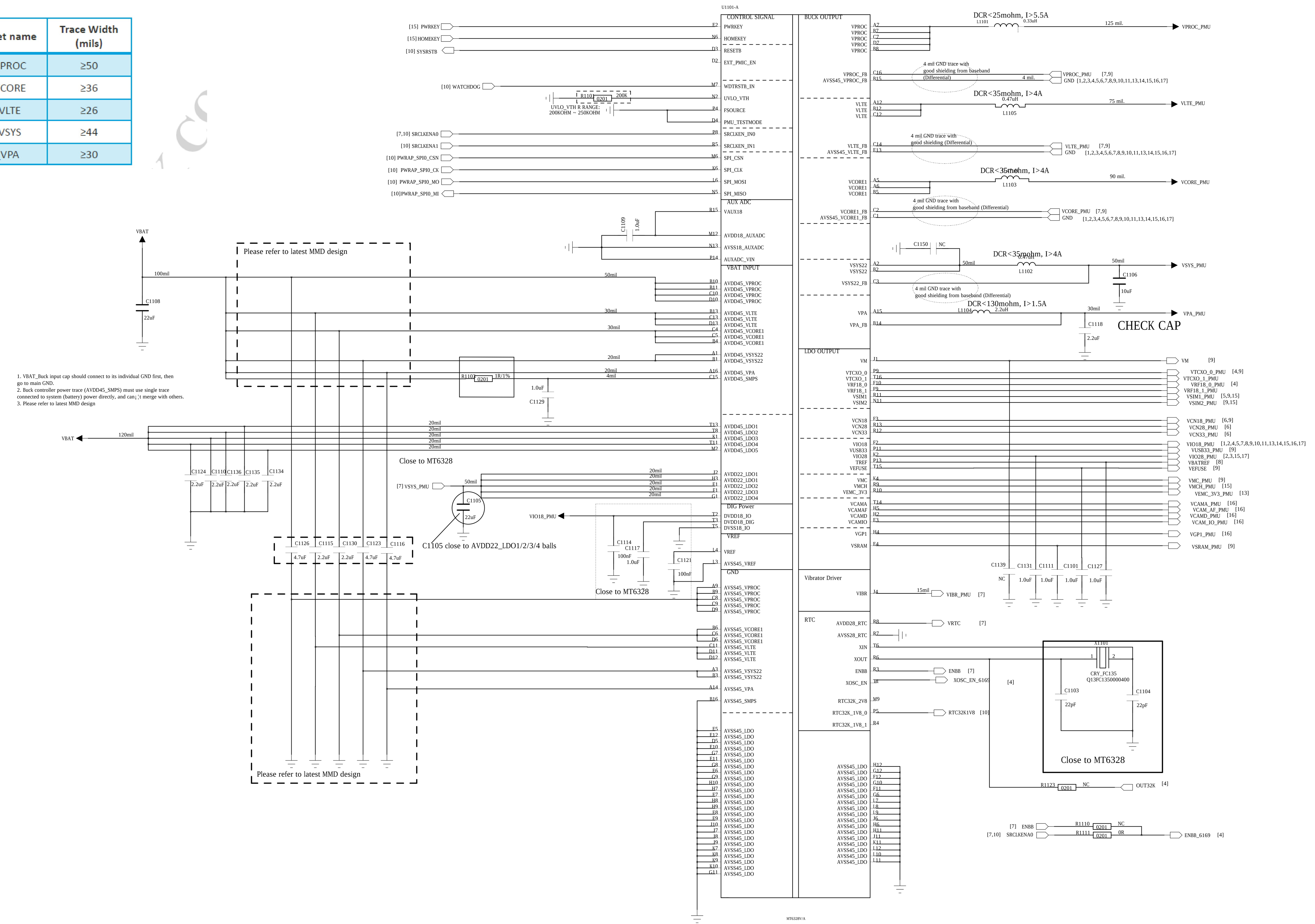
RELEASED: <Released By>	DATED: <Release Date>

CODE:	SIZE:	DRAWING NO:	REV:
<Code>	A0	<Drawing Number>	A1

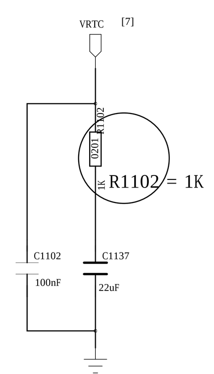
SCALE: <Scale> SHEET: 6 of 19

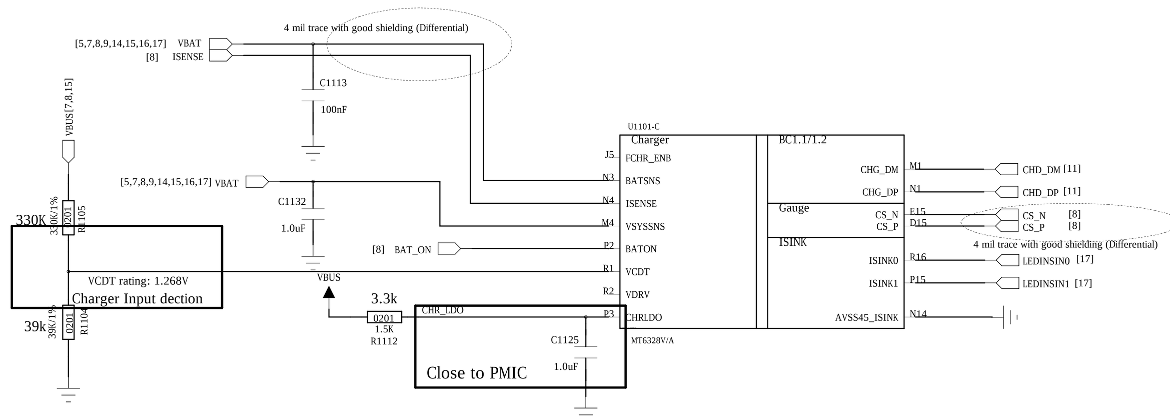
Place input cap. as close to PMICs as
VPROC>VCORE>VLTE>VSYS>VPA

Output net name	Trace Width (mils)
VBAT_VPROC	≥50
VBAT_VCORE	≥36
VBAT_VLTE	≥26
VBAT_VSYS	≥44
VBAT_VPA	≥30

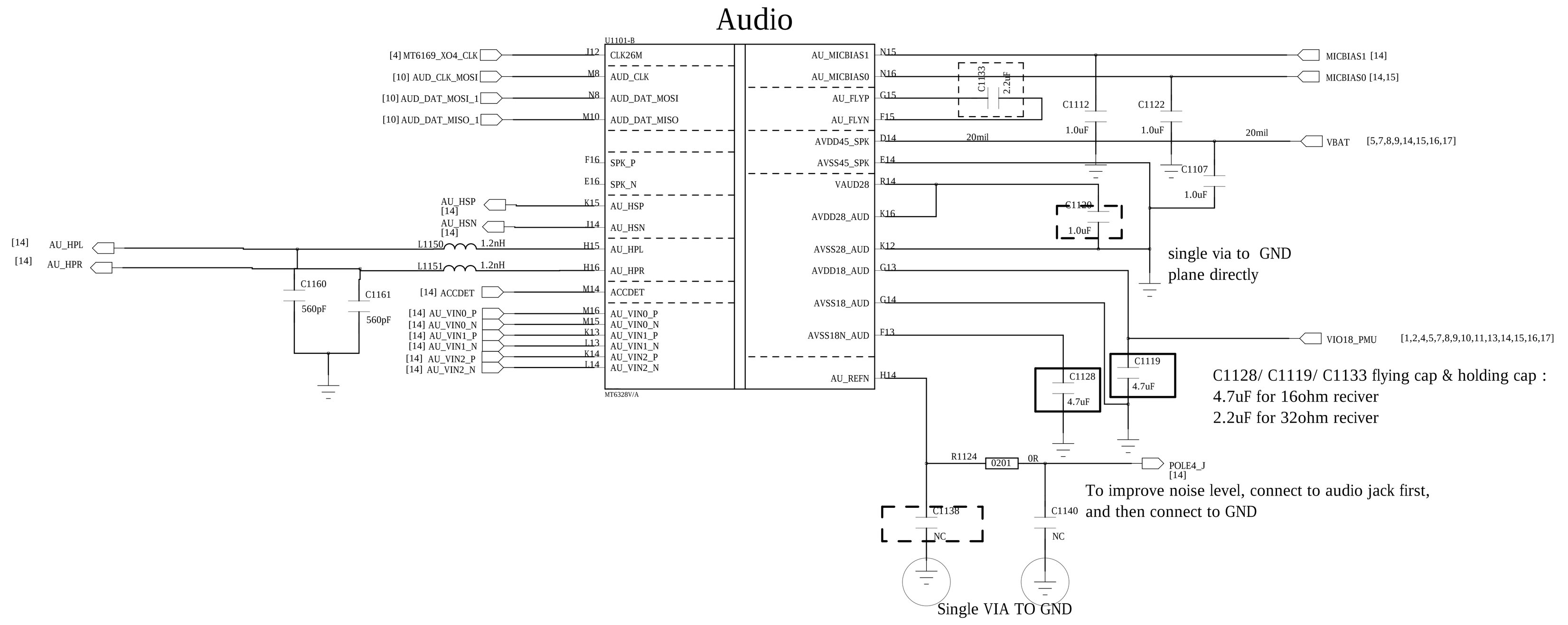


Regulator	Output Voltage Range(V)	Output Current(mA)	Input Decoupling Cap	Output Decoupling	Note
VPROC	0.6~1.31	5000	>2.2uF	0.33uH+47uF*2+22uF	≧125mil
VCORE	0.6~1.31	3500	>2.2uF	0.47uH+47uF*2	≧88mil
VLTE	0.6~1.31	2800	>2.2uF	0.47uH+22uF*3	≧63mil
VSYS	2	1900	>2.2uF	0.47uH+22uF+10uF	≧48mil
VPA	0.5~3.4	600	>4.7uF	2.2uH+2.2uF+2.2uF	≧30mil
VM	1.24/1.39/1.54	1000		9.4uF~10uF	L/W=1500mil/25mil
VTCXO_0	2.8	40		1uF~3uF	L/W=2800mil/6mil
VTCXO_1	2.8	40		1uF~3uF	L/W=2800mil/6mil
VRF18_0	1.825	350		2.2uF~ˇ3.3uF	L/W=2800mil/10mil
VRF18_1	1.2/1.3/1.5/1.825	300		2.2uF~ˇ3.3uF	L/W=2800mil/10mil
VSIM1	1.7/1.8/1.86/2.76/3.0/3.1	50		1uF~3uF	L/W=2800mil/6mil
VSIM2	1.7/1.8/1.86/2.76/3.0/3.1	50		1uF~3uF	L/W=2800mil/6mil
VCN18	1.8	150		1uF~3uF	L/W=2800mil/10mil
VCN28	2.8	40		1uF~3uF	L/W=2800mil/6mil
VCN33	3.0/3.1/3.2/3.3/3.4/3.5/3.6	350		4.7uF~5uF	L/W=2800mil/12mil
VIO18	1.8	600		2.2uF~14uF	L/W=2800mil/20mil
VUSB33	3.3	20		1uF~3uF	L/W=2800mil/6mil
VIO28	2.8	200		2.2uF~6.6uF	L/W=2800mil/10mil
VEFUSE	1.8/1.9/2.0/2.1/2.2	200		1uF~3uF	L/W=2800mil/10mil
VMC	1.8/2.9/3.0/3.3	200		1uF~4.7uF	L/W=2800mil/10mil
VMCH	2.9/3.0/3.3	800		4.7uF~7uF	L/W=2800mil/20mil
VEMC_3V3	2.9/3.0/3.3	400		4.7uF~7uF	L/W=2800mil/12mil
VCAMA	1.5/1.8/2.5/2.8	200		2uF~3.3uF	L/W=2800mil/10mil
VCAMAF	1.2/1.3/1.5/1.8/2.0/2.8/3.0/3.3	200		2uF~3.3uF	L/W=2800mil/10mil
VCAMD	0.9/1.0/1.1/1.22/1.3/1.5	500		4.4uF~13.2uF	L/W=2800mil/16mil
VCAMIO	1.2/1.3/1.5/1.8	200		1uF~3uF	L/W=2800mil/10mil
VGP1	1.2/1.3/1.5/1.8/2.5/2.8/3.0/3.3	200		1uF~3uF	L/W=2800mil/10mil
VSRAM	0.6~1.31	400		4.7uF~7uF	L/W=2800mil/12mil
VIBR	1.2/1.3/1.5/1.8/2.0/2.8/3.0/3.3	100		1uF~3uF	L/W=2800mil/8il
VAUX18	1.8	40		1uF~3uF	L/W=2800mil/6il
VAUD28	2.8	40		1uF~3uF	L/W=2800mil/6il
DVDD18_DIG	1.8	20		1uF	L/W=800mil/4il
VRTC	2.8	2		0.1uF	L/W=800mil/4il

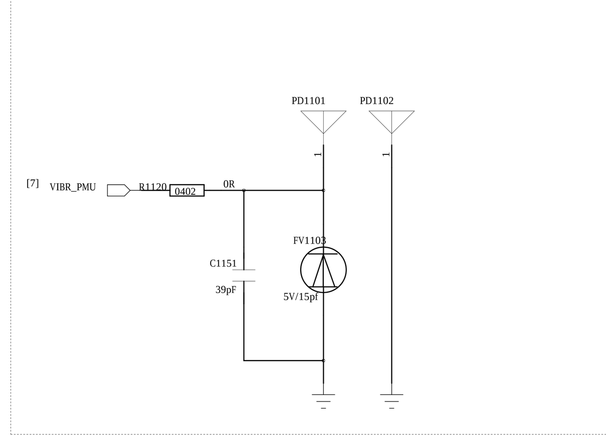




Audio



MOTO



Shanghai Longcheer

L3400

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CHECKED: <Checked By>	DATED: <Checked Date>				
QUALITY CONTROL: <QC By>	DATED: <QC Date>				
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>		SHEET: 19	

6 5 4 3 2 1

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

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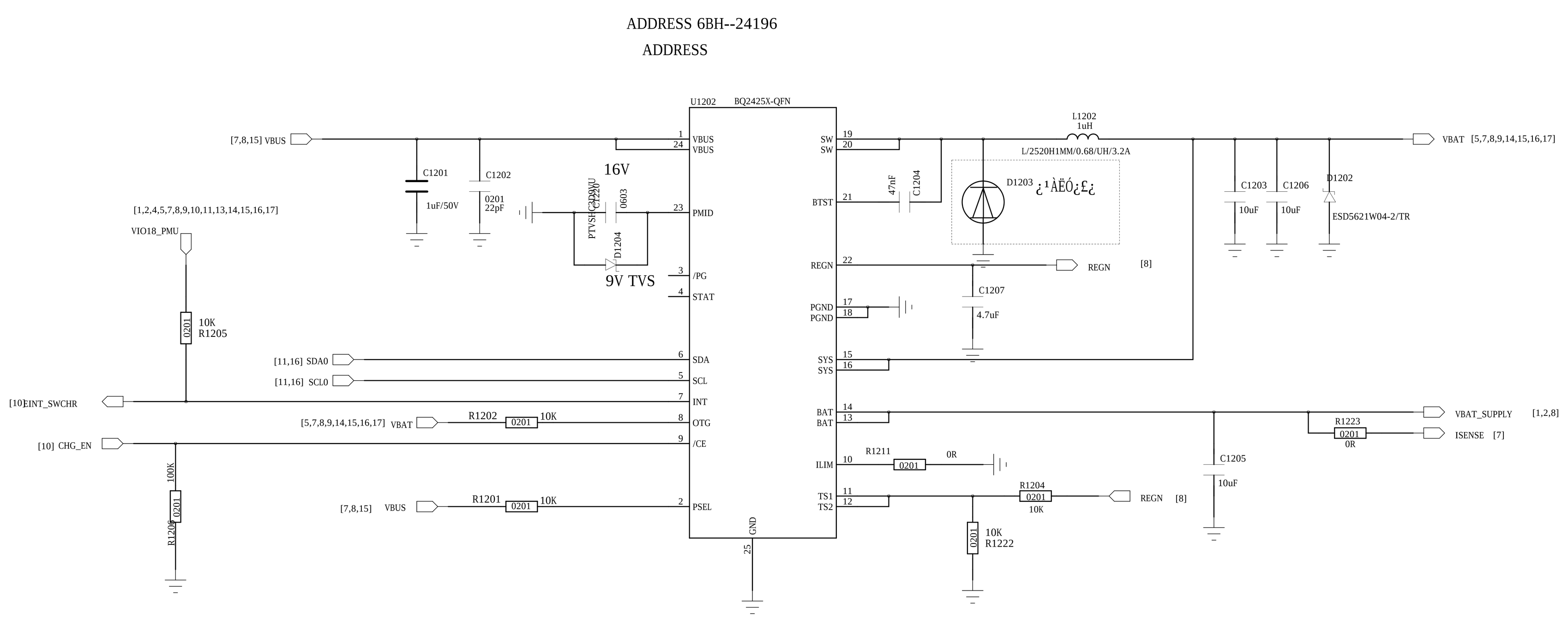

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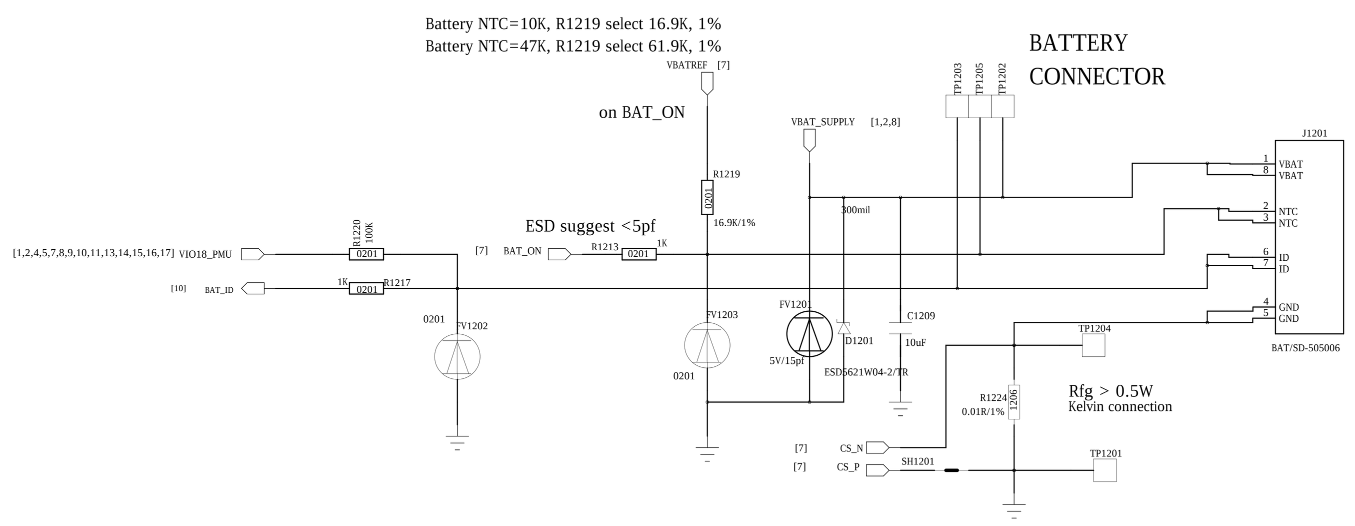

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COMPANY:	Shanghai Longcheer		
TITLE:	L3400		
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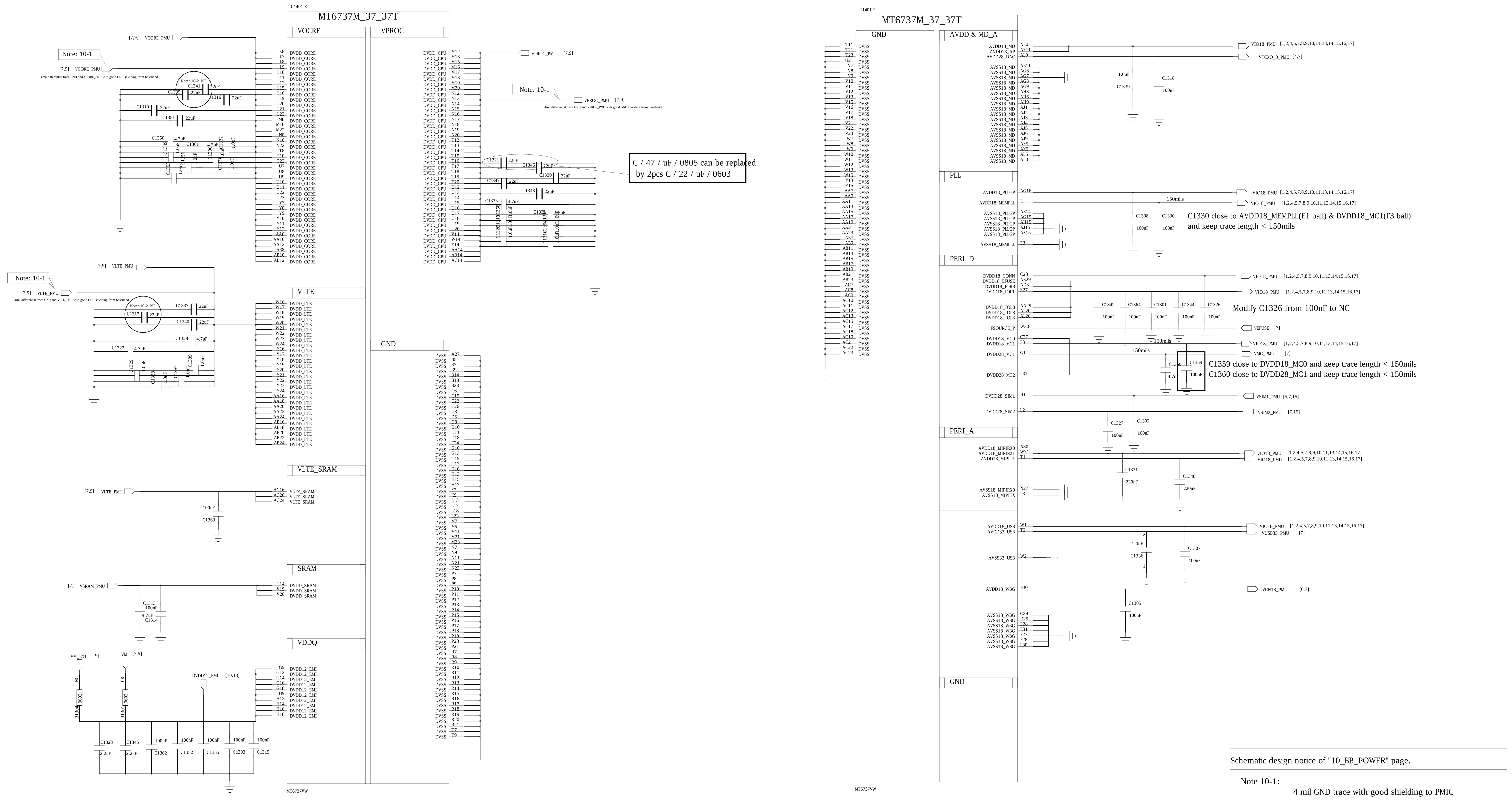


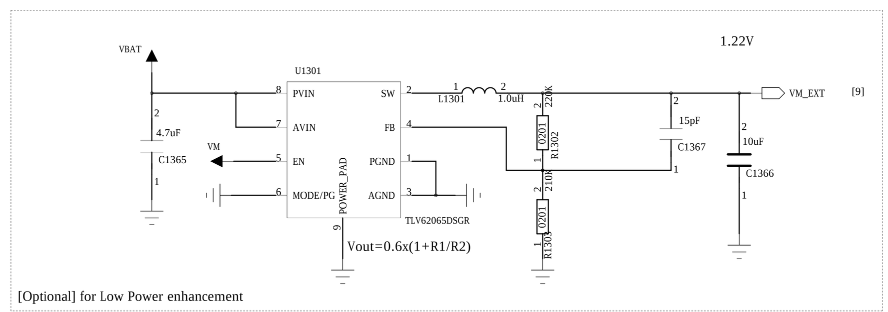
Schematic design notice of "10_BB_POWER" page.

Note 10-1: 4 mil GND trace with good shielding to PMIC (Differential)

Note 10-3:

FSOURCE_P(EFUSE)
(1)FSOURCE_P EFUSE power(VEFUSE) should be
only for EFUSE usage(not share with other application)
(2)W/I EFUSE program, VEFUSE need 1uF bypass cap
(pls refer to ¡§LDO output voltage/current table¡¨)
(3)W/O EFUSE program, VEFUSE bypass cap should be NC.

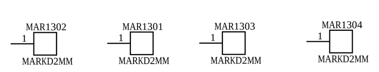
External Buck for DRAM



DRAM Buck Purpose: Reserved for low power performance

If using external buck for DRAM application, please refer to "MT6735_53 DRAM External Buck Design Notice":
1.Please add one more 1uF capacitor on PMIC VM output side (Must)
2.Please turn off Fast Transient Function of PMIC VM LDO (Must)

MARK POINT



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CHECKED: <Checked By>	DATED: <Checked Date>	TITLE: L3400		
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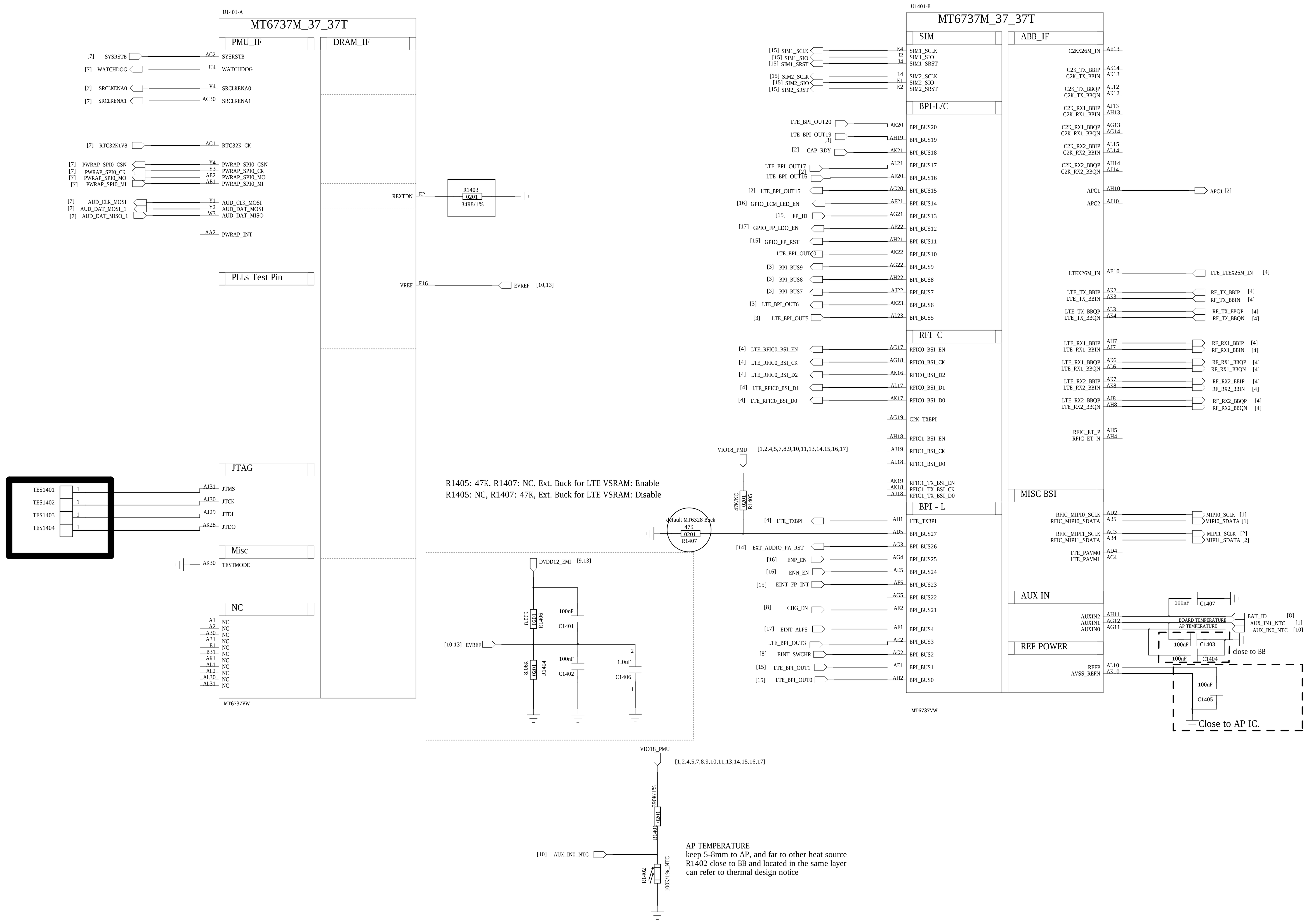

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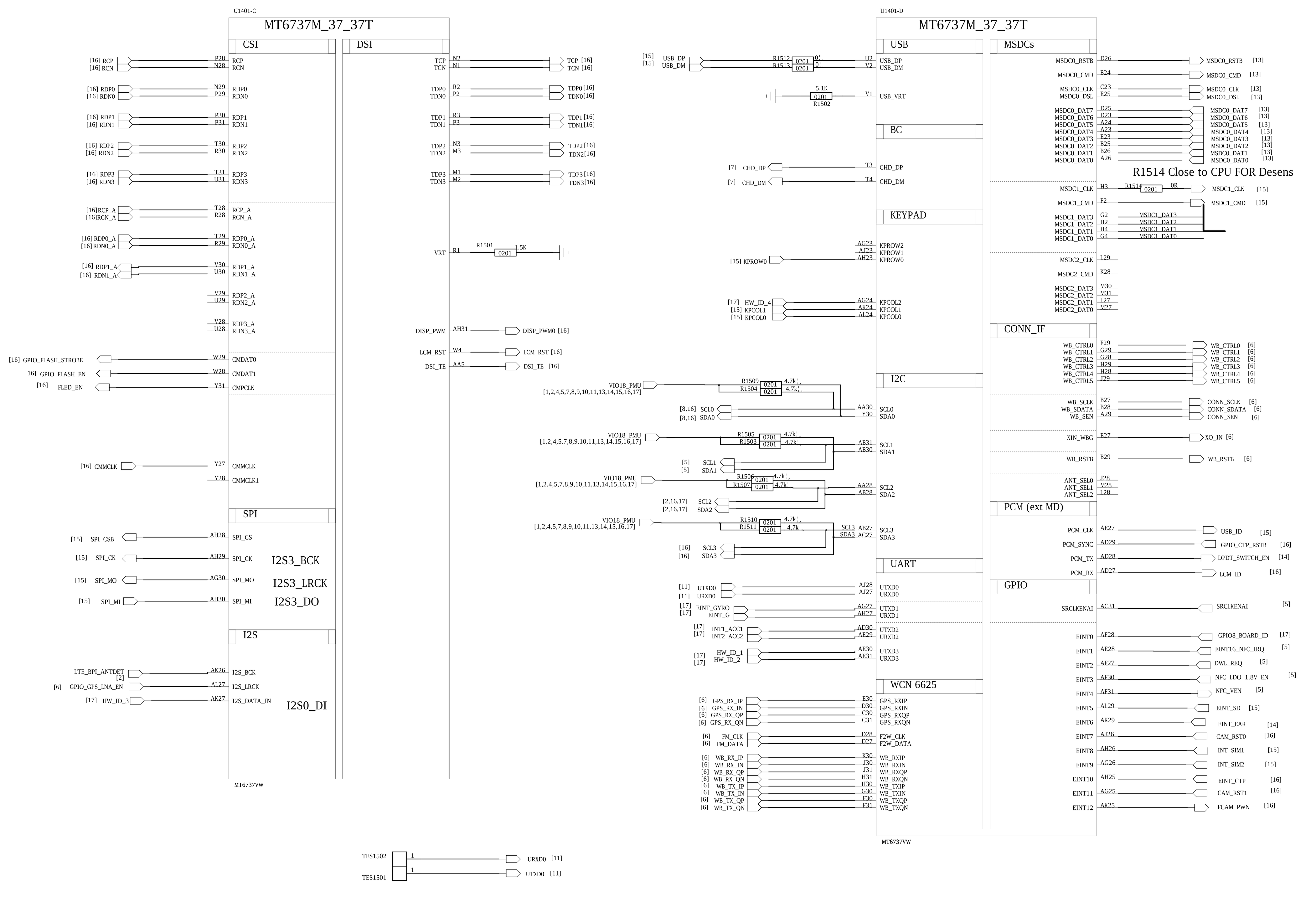

Schematic design notice of "12_BB_2" page.

Note 12-1: MSDC2 Pin should be NC (No Connection) for MT6735M

Note 12-2: MT6735M does Not have CMMCLK1 function

Please refer to MT6735M Design Notice for details

DRAWN: <Drawn By>	DATED: <Drawn Date>	COMPANY: Shanghai Longcheer		
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REVISION RECORD

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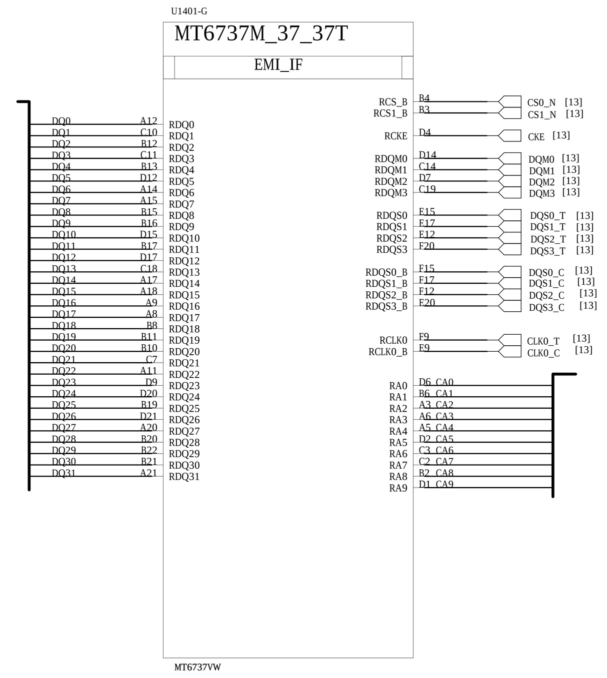

		COMPANY: Shanghai Longcheer			
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RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>		SHEET: 12 OF 19	

Connect to AP

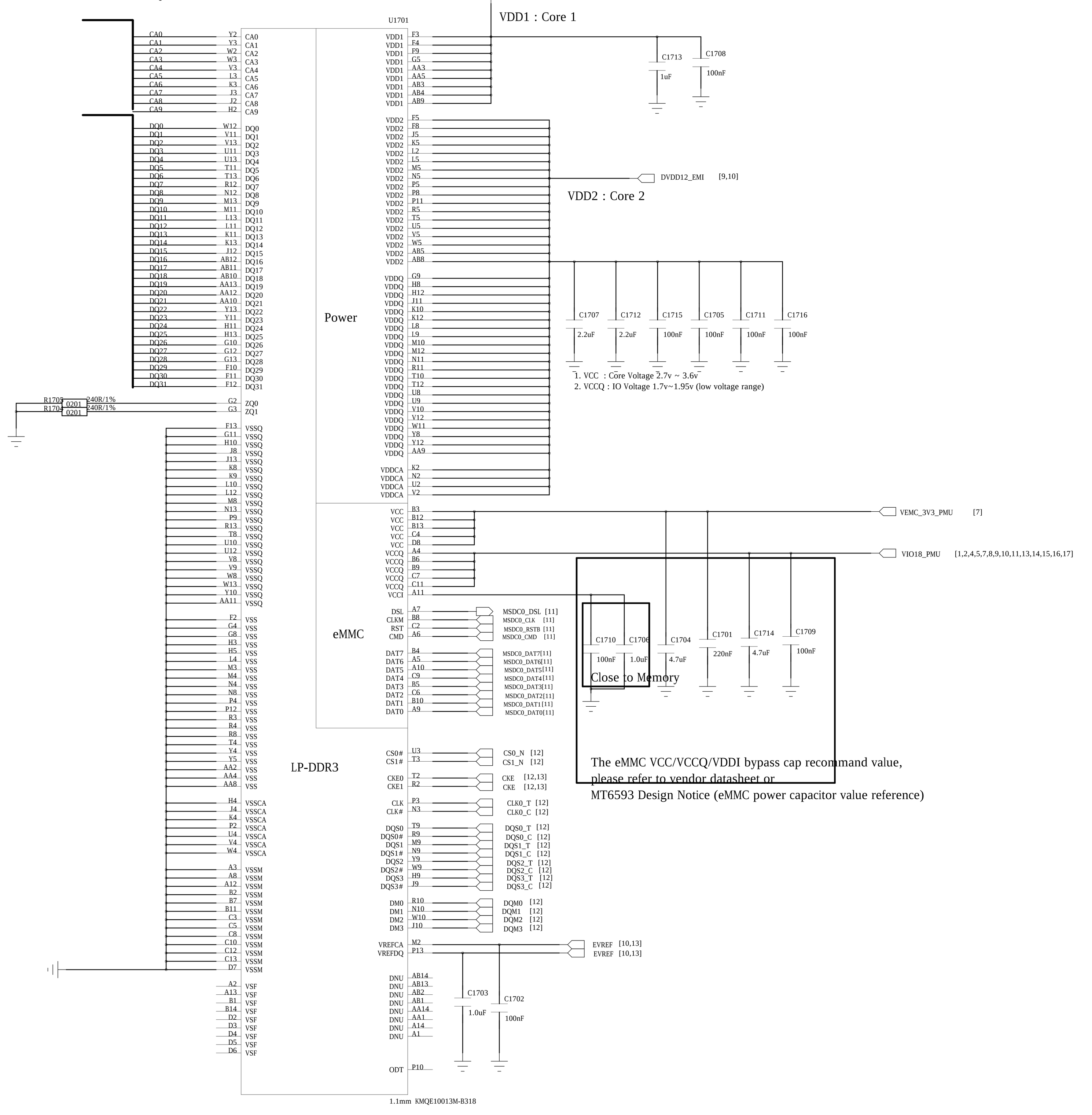
DQ[0..31]	DQ[0..31]
CA[0..9]	CA[0..9]
CS0_N	CS0_N
CS1_N	CS1_N
CKE	CKE
DQM0	DQM0
DQM1	DQM1
DQM2	DQM2
DQM3	DQM3
DQS0_C	DQS0_C
DQS1_C	DQS1_C
DQS2_C	DQS2_C
DQS3_C	DQS3_C
DQS0_T	DQS0_T
DQS1_T	DQS1_T
DQS2_T	DQS2_T
DQS3_T	DQS3_T
CLK0_T	CLK0_T
CLK0_C	CLK0_C
VREF_CA	VREF_CA
VREF_DQ	VREF_DQ
MSDC0_RSTB	MSDC0_RSTB
MSDC0_CMD	MSDC0_CMD
MSDC0_CLK	MSDC0_CLK
MSDC0_DSL	MSDC0_DSL
MSDC0_DAT0	MSDC0_DAT0
MSDC0_DAT1	MSDC0_DAT1
MSDC0_DAT2	MSDC0_DAT2
MSDC0_DAT3	MSDC0_DAT3
MSDC0_DAT4	MSDC0_DAT4
MSDC0_DAT5	MSDC0_DAT5
MSDC0_DAT6	MSDC0_DAT6
MSDC0_DAT7	MSDC0_DAT7

Power I/F

VIO18_PMU	VIO18_PMU
VEMC_3V3_PMU	VEMC_3V3_PMU
DVDD12_EMI	DVDD12_EMI

8GB eMMC + 8Gb LPDDR3
Hynix/H9TQ64A8GTMCUR-KUM

221 Ball, 0.5mm pitch
VDD1=1.8V
VDD2=1.20V
VDDCA=1.2V
VDDQ= 1.20V



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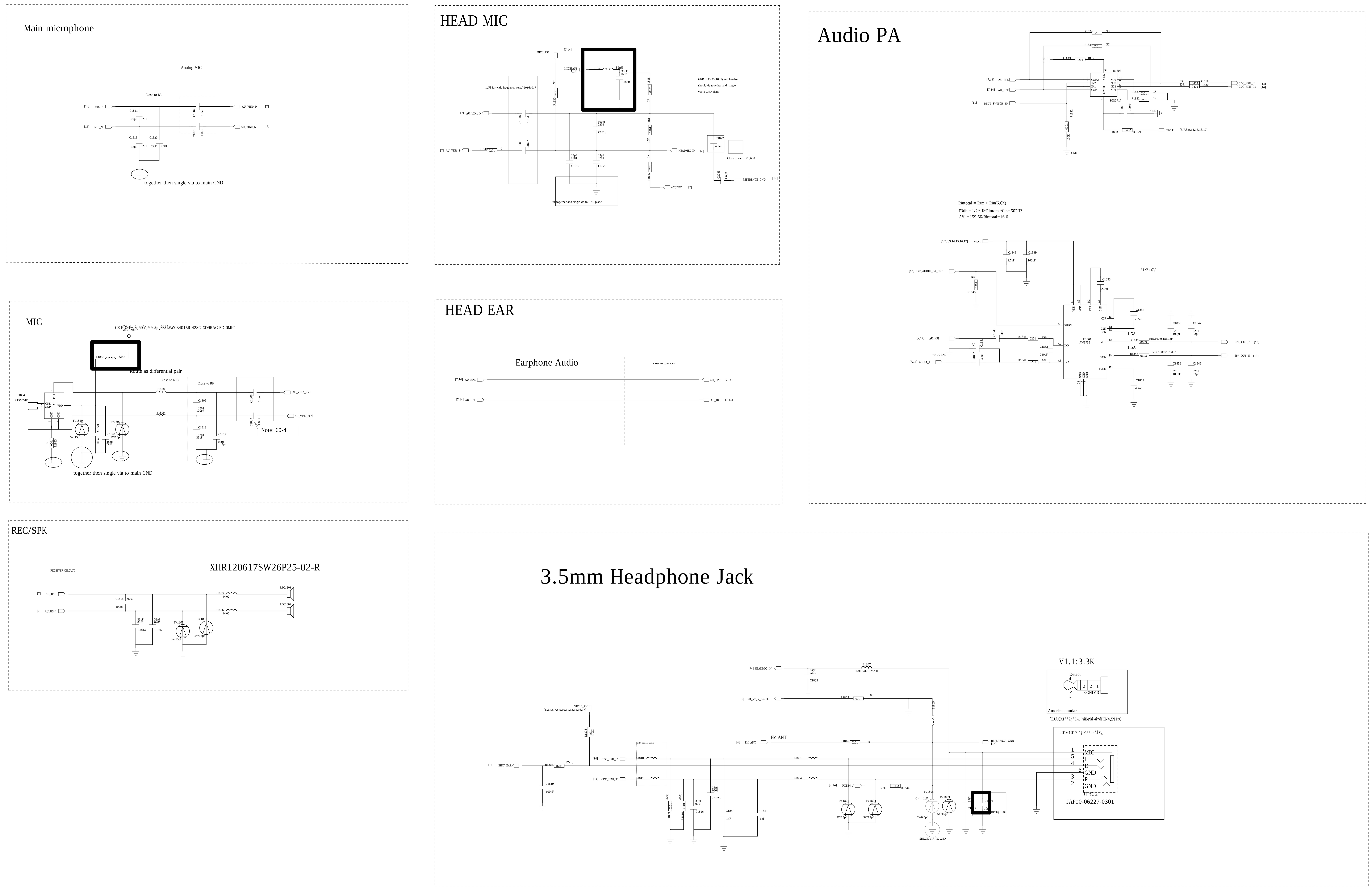

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RELEASED: <Released By>	DATED: <Release Date>		
CODE: <Code>	SIZE: A0	DRAWING NO: <Drawing Number>	REV: A1
SCALE: <Scale>	SHEET: 14 19		


Sub Connector
NANOSIM1
NANO SIM+MICRO SD CARD
SideKey+PowerKey
Signal | Description
HOMEKEY+GND | Volume Up
KPROW0+KPCOL0 | Volume Down
KYPD_PWR_N | PWER_ON
PM_RST_N + KYPD_PWR_N | Hardware Reset
vol+ = homekey + gnd
vol- = kpcol0 + kprow0
TOP
SPRING FOR GND
1.1mm
BOT
SPRING FOR GND
3.5mm
3.2mm
SHEILDING¡ª¡ªTOP
SHEILDING-BOT
R1923=0 R1924=NC
COMPANY: Shanghai Longcheer
TITLE: L3400
DRAWN: <Drawn By>
DATED: <Drawn Date>
CHECKED: <Checked By>
DATED: <Checked Date>
QUALITY CONTROL: <QC By>
DATED: <QC Date>
RELEASED: <Released By>
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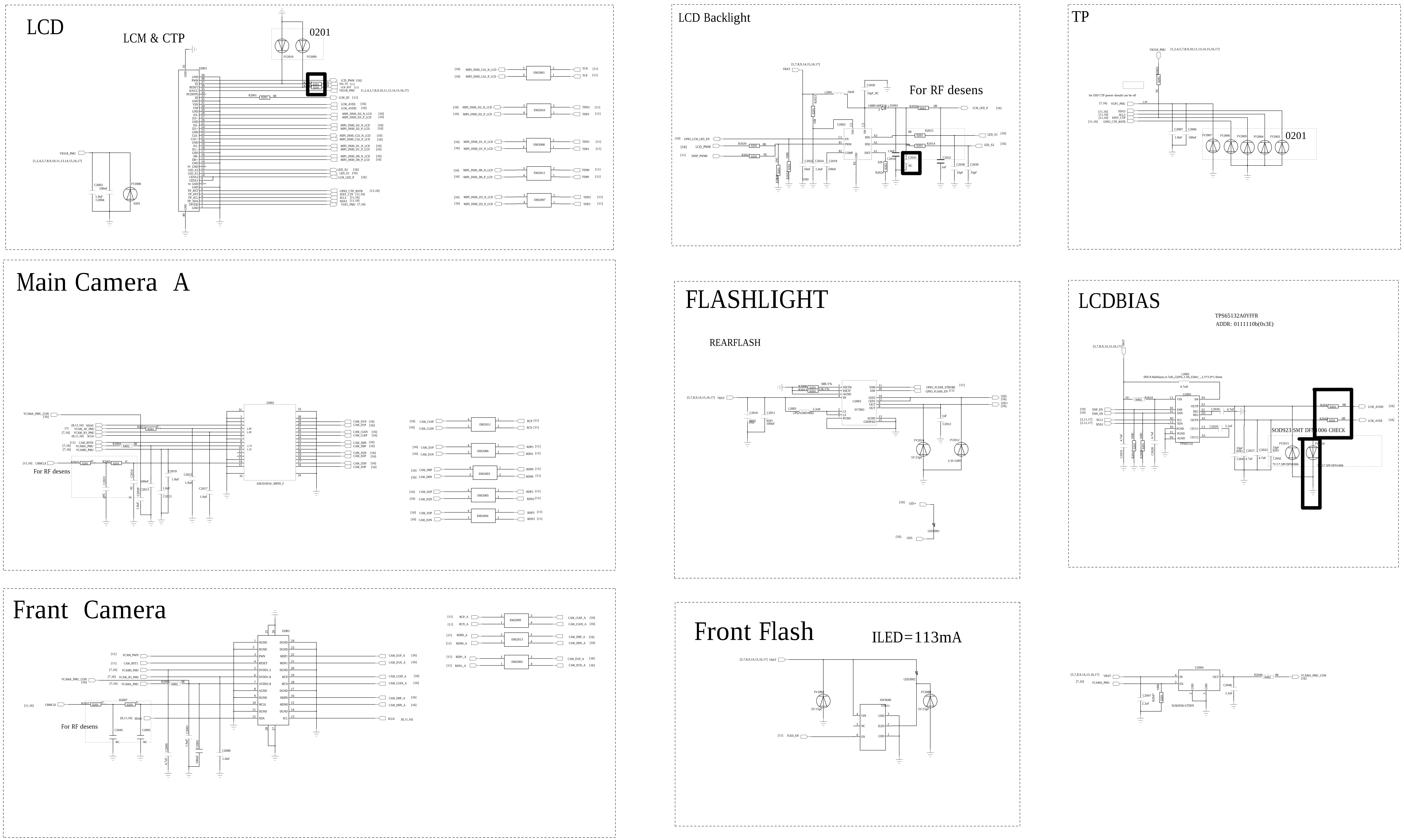
SHEET: 16 19



COMPANY: Shanghai Longcheer				
TITLE: L3400				
DRAWN: <Drawn By>	DATED: <Drawn Date>			
CHECKED: <Checked By>	DATED: <Checked Date>	CODE: <Code>	SIZE: A0	DRAWING NO: <Drawing Number>
QUALITY CONTROL: <QC By>	DATED: <QC Date>			REV: A1
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>	SHEET: 16 of 19	

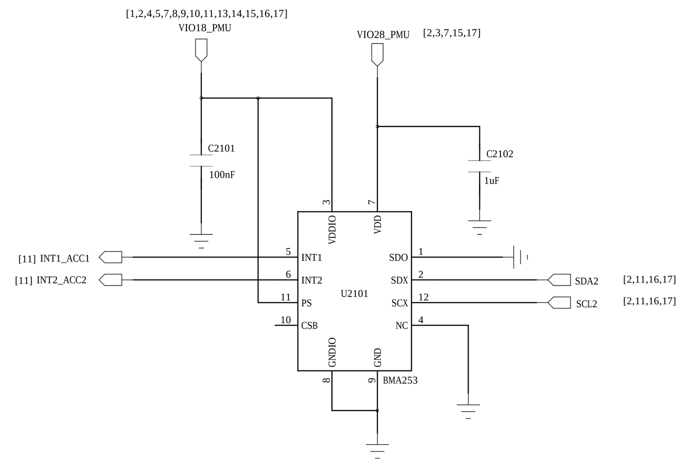
6	5	4	3	2	1

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

Accelerometer Sensor

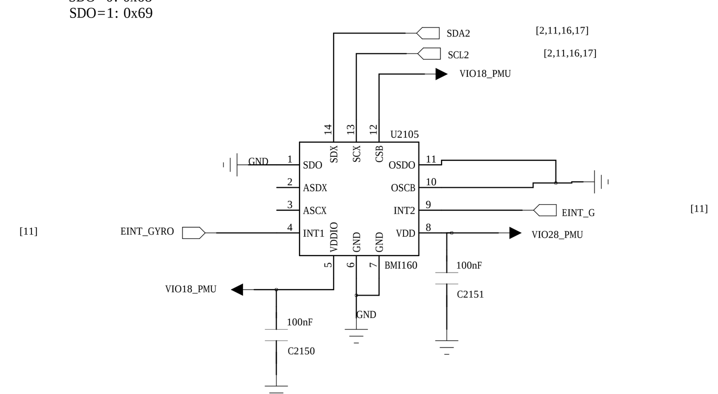
8bit(BMA223) compatible 12bit(BMA253)

ADDR 0011000b(0x18) SDO=0

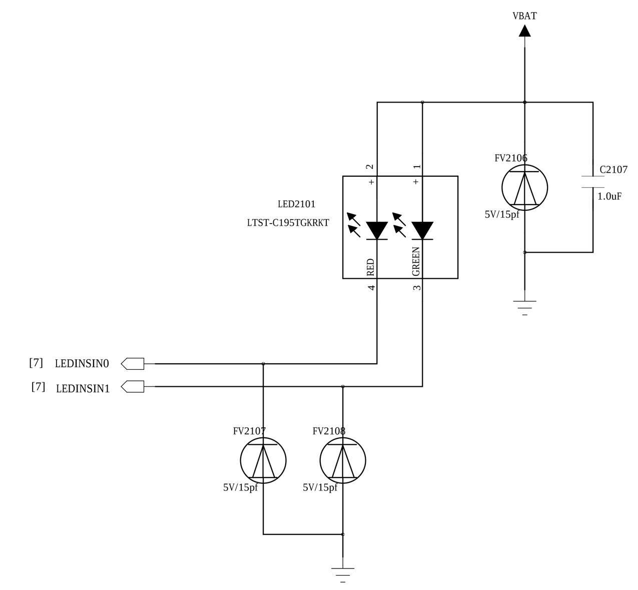


G+Gyro-Sensor

BMI120 I2C address
SDO=0: 0x68
SDO=1: 0x69



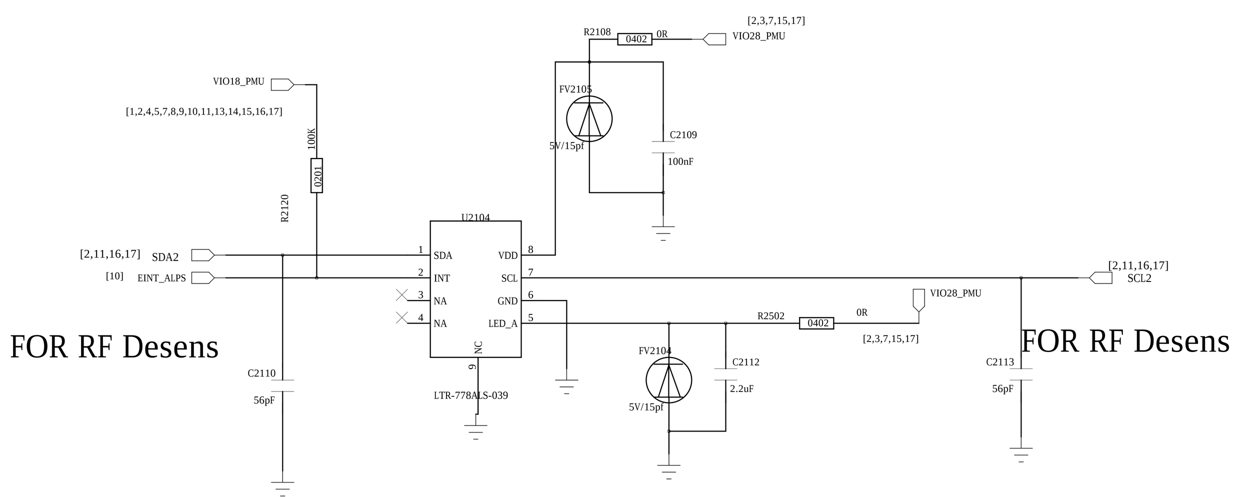
RG LED



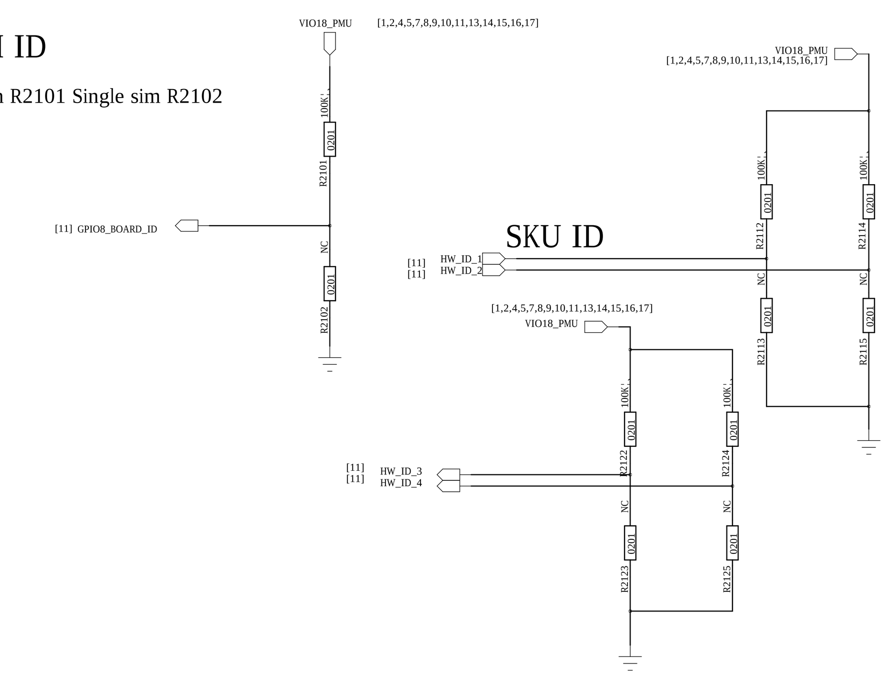
PS Sensor

EPL2590KTWJP

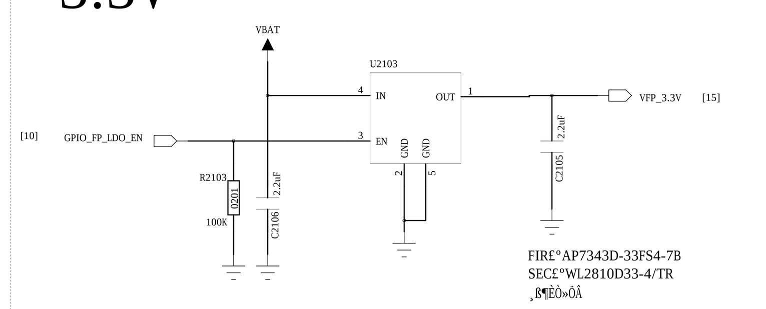
I2C address: 0xA6H Write 0xA7H Read



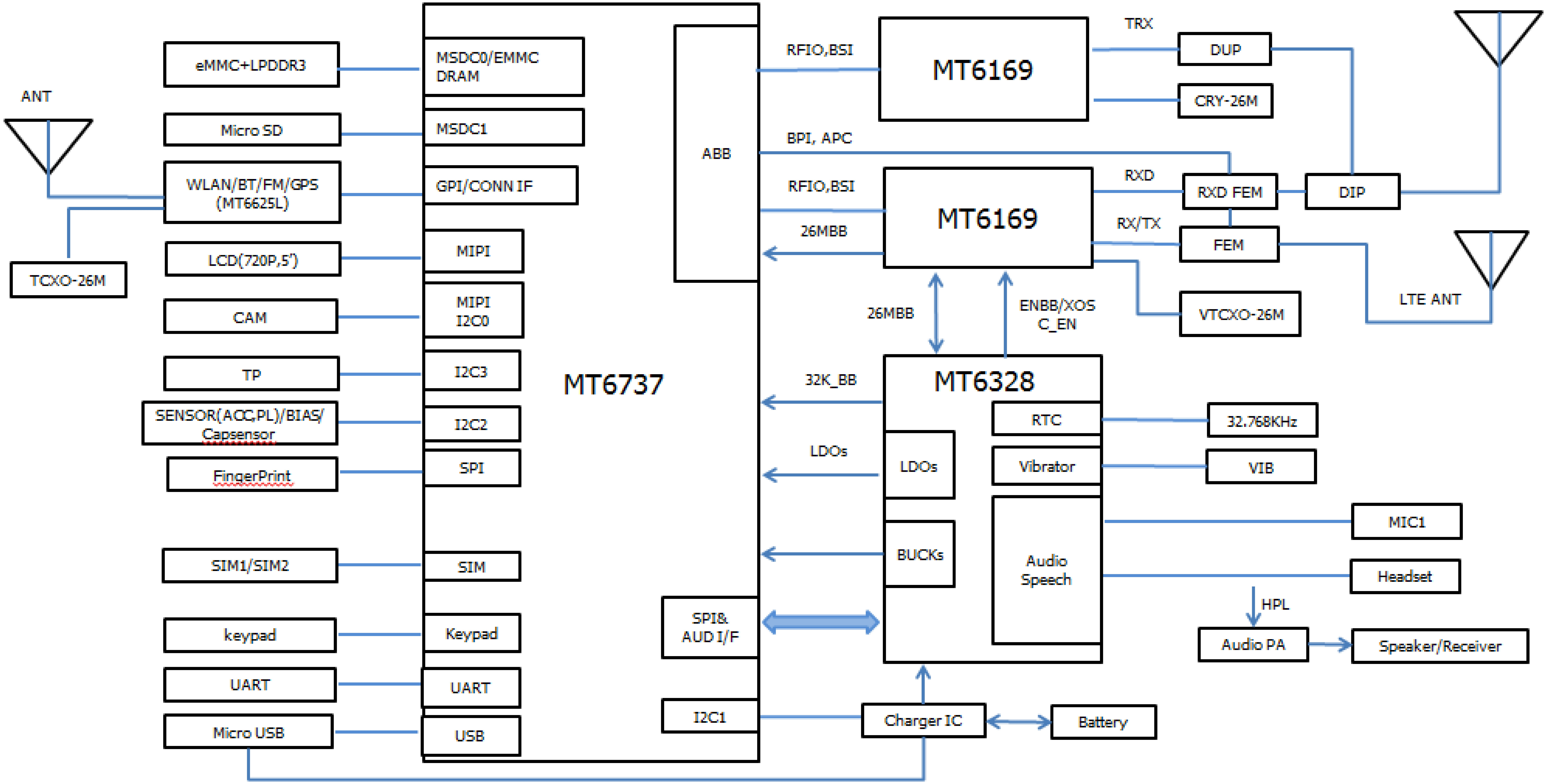
BORAD ID



FingerPrint-Sensor



COMPANY:	Shanghai Longcheer				
TITLE:	L3400				
DRAWN: <Drawn By>	DATED: <Drawn Date>	CODE:	SIZE:	DRAWING NO:	REV:
CHECKED: <Checked By>	DATED: <Checked Date>	<Code>	A0	<Drawing Number>	A1
QUALITY CONTROL: <QC By>	DATED: <QC Date>				
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>		SHEET: 17	19



Shanghai Longcheer

L3400

<Code>	A0	<Drawing Number>	A1

I2C	Function	I2C Spec.	Budget Timing	I2C Slave Address (7-bit mode)
I2C-0	Rear Camera - 5M	400 Kbps	Yes.	Rear camera (IMX135) I2C address: 0X10 (Write:0x20, Read:0x21) AF driver (DW9714A) I2C address: 0x0C (Write:0x18 , Read:0x19)
	Front Camera - 2M	400 Kbps	Yes.	Front camera (GC2355) I2C address: 0x3C (Write:0x78, Read:0x79)
I2C-1	CTP	400 Kbps	Yes.	GT1151 / CTP I2C address: 0X5D (Write:0xBA, Read:0xBB) or 0x14 (Write:0x28, Read:0x29)
I2C-2	M Sensor	2.5 Mbps		BM223 / M-Sensor I2C Address: 0x30
	Gyro Sensor	400 Kbps	Yes.	
	Accelerometer	400 Kbps		
	A/PS Sensor	400 Kbps		LTR-553ALS-WA / AS+PS I2C address: 0x46
	NFC	1.2 Mbps	Yes.	
	Flashlight Driver	400 Kbps		
I2C-3				

Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

COMPANY:	Shanghai Longcheer
TITLE:	L3400

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

CODE:	SIZE:	DRAWING NO:	REV:
<Code>	A0	<Drawing Number>	A1

SCALE: <Scale> SHEET: 19 OF 19